upGrad ENTERPRISE

Working with Views and Indexes in SQL Server

By

Narasimha Rao T

Microsoft.Net FSD Trainer

Professional Development Trainer

tnrao.trainer@gmail.com

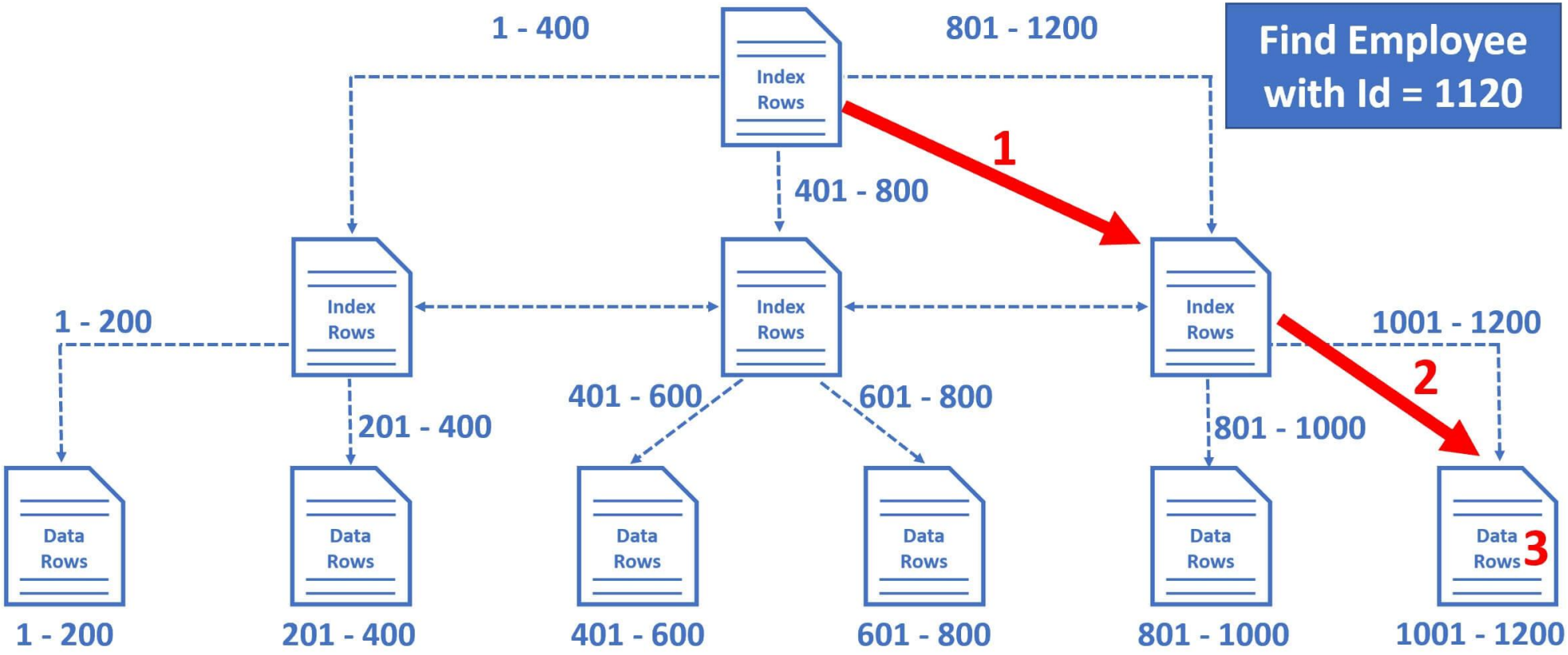
Microsoft® SQL Server

Day-20 (05-Aug-2025)

Working with Indexes

Importance of Indexes

- Indexes **speed up data retrieval**
- Work like a book index — quickly find what you need


Find Employee with Id = 1120
1 - 400
801 - 1200
Index Rows
401 - 800
1
1 - 200
Index Rows
Index Rows
Index Rows
1001 - 1200
2
401 - 600
601 - 800
201 - 400
801 - 1000
Data Rows
Data Rows
Data Rows
Data Rows
Data Rows
Data Rows
3
1 - 200
201 - 400
401 - 600
601 - 800
801 - 1000
1001 - 1200

Types of Indexes

1. Clustered Index

- One per table, sorts table data

```
CREATE CLUSTERED INDEX idx_EmpID ON Employees(EmpID);
```

2. Non-Clustered Index

- Separate structure with pointers

```
CREATE NONCLUSTERED INDEX idx_LastName ON Employees(LastName);
```

3. Unique Index

- Ensures column(s) have unique values

```
CREATE UNIQUE INDEX idx_Email ON Employees(Email);
```

4. Composite Index

- Includes multiple columns

```
CREATE INDEX idx_NameDept ON Employees(LastName, Department);
```

Index Management

CREATE INDEX

```
CREATE INDEX idx_Salary ON Employees(Salary);
```

DROP INDEX

```
DROP INDEX idx_Salary ON Employees;
```

ALTER INDEX

Rebuild (fixes fragmentation):

```
ALTER INDEX ALL ON Employees REBUILD;
```

Reorganize (less intensive):

```
ALTER INDEX ALL ON Employees REORGANIZE;
```

Disable:

```
ALTER INDEX idx_Salary ON Employees DISABLE;
```

Included Columns & Covering Indexes

Covering Index: An index that contains *all* the columns required by a query.

```
CREATE NONCLUSTERED INDEX idx_Covering
ON Employees(DepartmentID)
INCLUDE (FirstName, LastName);
```

Filtered Indexes

- Index with a WHERE clause

```
CREATE NONCLUSTERED INDEX idx_ActiveEmployees
ON Employees(IsActive)
WHERE IsActive = 1;
```

Case Study:

E-Commerce App:

- Use filtered index for `Orders WHERE Status = 'Pending'`
- Clustered index on `OrderID`
- Composite index on `(CustomerID, OrderDate)` for dashboard reports

1. VIEWS in SQL Server

What is a View?

A *View* is a virtual table based on the result of a SQL query.

CREATE VIEW

```
CREATE VIEW vw_EmployeeDetails AS
SELECT EmpID, FirstName, LastName, Department
FROM Employees
WHERE IsActive = 1;
```

Querying a View

```
SELECT * FROM vw_EmployeeDetails;
```

Benefits of Views

- **Security**: Limit data access to certain users
- **Simplicity**: Hide complex joins
- **Abstraction**: Decouple schema from application logic

ALTER VIEW

```sql
ALTER VIEW vw_EmployeeDetails AS
SELECT EmpID, FirstName, LastName, Department, Salary
FROM Employees
WHERE IsActive = 1;
```

DROP VIEW

```sql
DROP VIEW vw_EmployeeDetails;
```